DATA SCIENCE & ARTIFICIAL INTELLIGENCE & CS/IT



Calculus and Optimization

Limit Continuity and differentiablity

LEC - 03



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Recap of previous lecture



Topic

BASICS of LIMITS & CONTINUITY

(PART-2)

Topics to be Covered



Topic

BASICS of LIMITS & CONTINUITY

(Part-3)

RECAP

① Infinity → Concept which is Beyond Imagination

& $\frac{\text{Something}}{\infty} \cong 0$, $\frac{\text{Something}}{0} = \text{N.D}$ (either ∞, or $-\infty$)

② Indeterminate form → If Mathematical expression has Multiple Ans.

eg $\frac{0}{0}, \frac{\infty}{\infty}, 0\times\infty, \infty-\infty, 0^0, \infty^0, 1^\infty$

③ N.D form → $\infty+\infty, \infty\times\infty, \infty^\infty, \frac{\infty}{\infty}, -\infty$ etc.

④ limit exist ⟹ It is possible to find Unique, Constant Value that must be free form x (i.e. LHL = RHL = Unique Const Value.)

(Approx Value)

RECAP.

(5) Nbd → Nbd of 'a' = $(a-h, a+h)$ where $h \to 0$ i.e. h is very small +ve Numbers.

If x lies in the Nbd of 'a' $\Rightarrow x \in (a-h, a+h)$ or $a-h < x < a+h$ or $x \to a$

eg x " " " 5 $\Rightarrow x \in (4.999, 5.001)$ i.e. $h = 0.001$

(6) L-Hospital Rule → Applicable only for $\frac{0}{0}$ & $\frac{\infty}{\infty}$ form. $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

Separately Diff N^r & D^r equal no of times until we are free form.

eg x lies in the Nbd of $-5 \Rightarrow x \in (-5.001, -4.999)$, $h = 0.001$

(7) $\lim_{x \to a} f(x) = l \Rightarrow$ whenever x lies in the Nbd of a (i.e. on x axis) $f(x)$ . . . . . l (i.e. on y axis)

(8) Continuity: If $\lim_{x \to a} f(x) = f(a) \Rightarrow f(x)$ is Cont at $x = a$

i.e. Approx Value = Exact Value i.e. $(LHL = RHL) = f(a)$

RECAP.

Sp. Note: To check Discontinuity of $f(x)$ at $x=a$ we have 3 Methods;

(M-I) If Exact Value = DNE then $f(x)$ is Discontinuous

(M-II) if Approx Value = DNE then " " "

(M-III) if Both exist But not equal then $f(x)$ is Discont.

i.e. $\lim_{x \to a} f(x) \neq f(a) \Rightarrow f(x)$ is Discont at $x=a$

Method of Solving Questions →

(M-I) By Direct Substitution (Best Method).

(M-II) By factorisation

(M-III) By Rationalisation

(M-IV) By using IND form Concept

(M-V) By using Standard Results

Sp. formulaes :-

(1) $\sum_{N=1}^{N} N = 1+2+3+\ldots+N = \frac{N(N+1)}{2}$

(2) $\sum_{N=1}^{N} N^2 = 1^2+2^2+3^2+\ldots+N^2 = \frac{N(N+1)(2N+1)}{6}$

(3) $\sum_{N=1}^{N} N^3 = 1^3+2^3+3^3+\ldots+N^3 = \left(\frac{N(N+1)}{2}\right)^2$

(4) $\sum_{N=1}^{N} (a) = \underbrace{a+a+a+\ldots+a}_{N \text{ times}} = N \cdot a$

Q1. $\lim_{x\to 2}\left(\frac{x^2-4}{x+3}\right)=?=\frac{4-4}{2+3}=\frac{0}{5}=0$ i.e. unique so limit exist.

Q2. $\lim_{n\to\infty}\left(\frac{1+2+3+4+\cdots+n}{n^2}\right)=?=\lim_{n\to\infty}\left[\frac{\frac{n(n+1)}{2}}{n^2}\right]=\frac{1}{2}\lim_{n\to\infty}\frac{n^2\left(1+\frac{1}{n}\right)}{n^2}$

$=\frac{1}{2}\lim_{n\to\infty}\left(1+\frac{1}{n}\right)=\frac{1}{2}\left(1+\frac{1}{\infty}\right)=\frac{1}{2}(1+0)=\frac{1}{2}$

Q3. $\lim_{n\to\infty}\left[\frac{1^2+2^2+3^2+\cdots+n^2}{n^3}\right]=?=\lim_{n\to\infty}\left[\frac{\frac{n(n+1)(2n+1)}{6}}{n^3}\right]$

$=\lim_{n\to\infty}\frac{n^3\left\{\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right\}}{6.n^3}=\frac{(1+0)(2+0)}{6}=\frac{1}{3}$ Ans

Q. $\lim_{n\to\infty}\left[\frac{1-2+3-4+5-6+\ldots\ldots+(2n-1)-2n}{\sqrt{n^2+1}+\sqrt{n^2-1}}\right]=?$

(a) 1/2

(b) -1/2

(c) 1

(d) D.N.E

$$=\lim_{n\to\infty}\left[\frac{(1-2)+(3-4)+(5-6)+\ldots\ldots+((2n-1)-2n)}{n\sqrt{1+\frac{1}{n^2}}+n\sqrt{1-\frac{1}{n^2}}}\right]$$

$$=\lim_{n\to\infty}\left\{\frac{\overbrace{(-1)+(-1)+(-1)+\ldots\ldots+(-1)}^{n\text{ times}}}{n\left[\sqrt{1+\frac{1}{n^2}}+\sqrt{1-\frac{1}{n^2}}\right]}\right\}$$

$$=\lim_{n\to\infty}\left[\frac{-n}{n\left\{\sqrt{1+\frac{1}{n^2}}+\sqrt{1-\frac{1}{n^2}}\right\}}\right]=\frac{-1}{\sqrt{1+0}+\sqrt{1-0}}=\frac{-1}{2}$$

Q① $\lim_{x \to 0} \left(\frac{1}{x}\right) = \frac{1}{0} = (+\infty \text{ or } -\infty)$ i.e. ND $\Rightarrow$ limit DNE



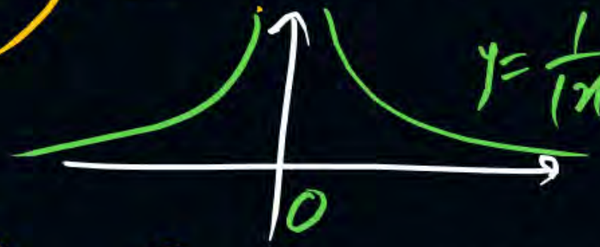
Q② $\lim_{x \to +\infty} \left(\frac{1}{x}\right) = \frac{1}{\infty} = 0$ i.e. unique = exist

Q③ $\lim_{x \to 0} \left(\frac{1}{x^2}\right) = \frac{1}{0^2} = +\infty \Rightarrow$ $\because \infty$ is Not unique $\Rightarrow$ limit DNE



Q④ $\lim_{x \to +\infty} \left(\frac{1}{x^2}\right) = \frac{1}{\infty^2} = \frac{1}{\infty} = 0 \Rightarrow$ exist.

Q. $\lim_{x \to 0} \left(\frac{1}{|x|}\right) = ? = \infty \Rightarrow$ DNE



Q. $\lim_{x \to \infty} \left(\frac{1}{|x|}\right) = ? = 0 \Rightarrow$ Exist

Note:

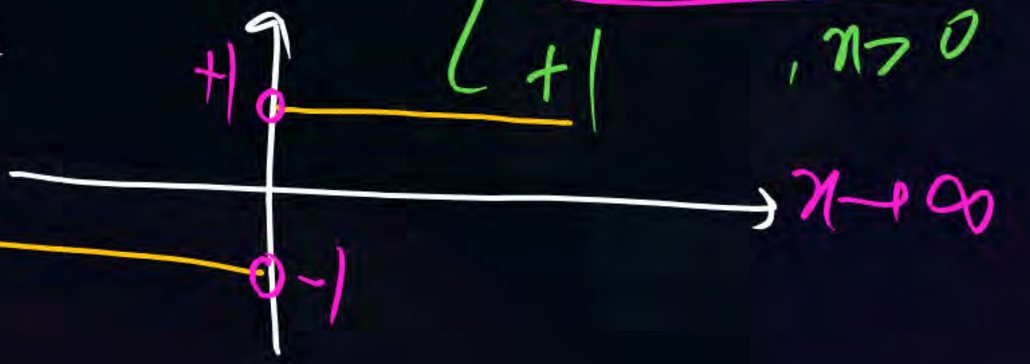
$$\text{Sgn}(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} -1, & x < 0 \\ 1, & x > 0 \\ 0, & x = 0 \end{cases}$$

Q. $\lim_{x \to 0} \left(\frac{|x|}{x}\right) = ?$ → LHL $= -1$, RHL $= +1$; i.e limit in the Nbd of $x = 0$ DNE

$$f(x) = \frac{|x|}{x} = \begin{cases} -1, & x < 0 \\ \text{DNE}, & x = 0 \\ +1, & x > 0 \end{cases}$$

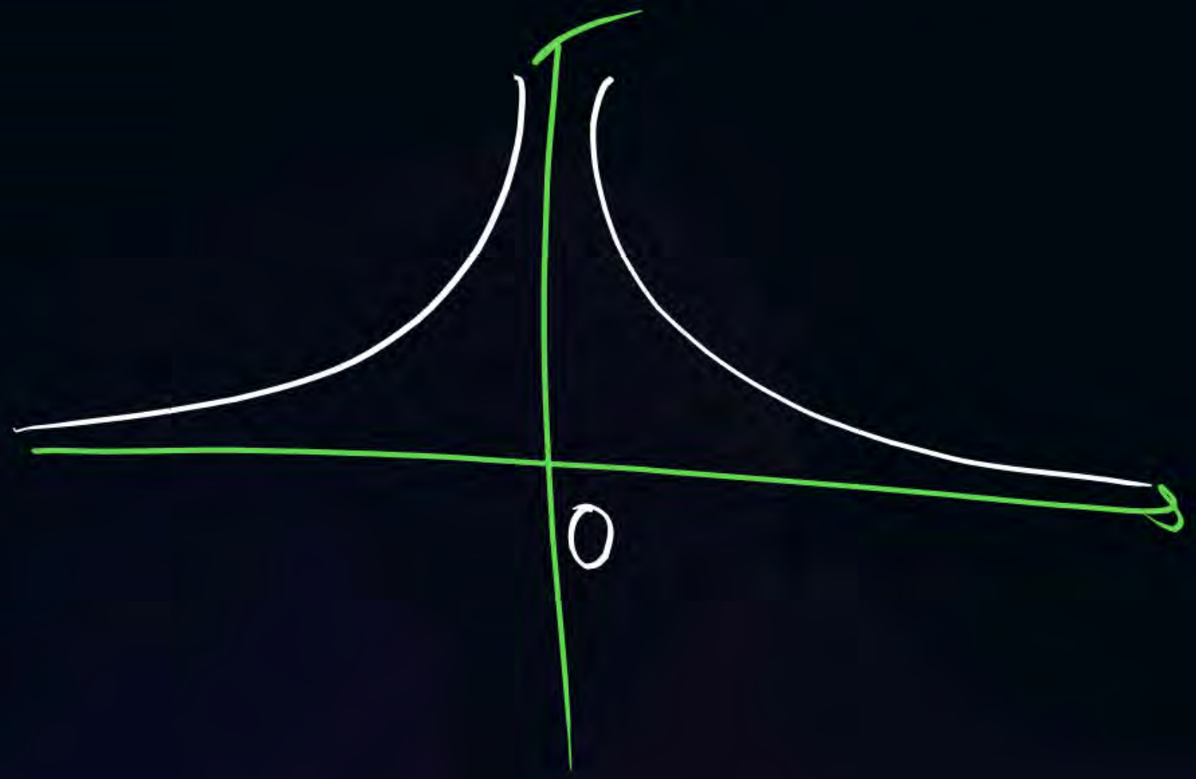
Q. $\lim_{x \to \infty} \left(\frac{|x|}{x}\right) = ? = +1$ is exist

$\lim_{x \to 5} \left(\frac{|x|}{x}\right) = +1$, $\lim_{x \to -3} \left(\frac{|x|}{x}\right) = -1$



Q Draw Graph of

$$y = \frac{1}{|x|} = \begin{cases} \frac{-1}{x}, & x < 0 \\ +\frac{1}{x}, & x > 0 \end{cases}$$



Some Useful Result :-

$$a^3+b^3 = (a+b)(a^2+b^2-ab)$$

$$a^3-b^3 = (a-b)(a^2+b^2+ab)$$

$$(a+b)^3 = a^3+b^3+3ab(a+b)$$

$$(a-b)^3 = a^3-b^3-3ab(a-b)$$

$$a^2-b^2 = (a-b)(a+b)$$

Type (2): FACTORISATION

Q. $\lim_{x \to 2} \left(\frac{x^2 - 5x + 6}{x^2 - 4} \right) = ?$

$$= \lim_{x \to 2} \frac{(x-2)(x-3)}{(x-2)(x+2)}$$

$$= \lim_{x \to 2} \left(\frac{x-3}{x+2} \right) = \frac{2-3}{2+2} = \frac{-1}{4}$$

Note: $x \to 2 \Rightarrow (x-2) \to 0$

i.e. we can take $x - 2 = 0.001$

Q. $\lim_{y \to 1} \left(\frac{y^3 - 1}{y - 1} \right) = ?$

$$= \lim_{y \to 1} \left[\frac{(y-1)(y^2 + 1 + y)}{(y-1)} \right] = 1 + 1 + 1 = 3$$

Q. $\lim_{y \to 1} \left(\frac{y^{\frac{1}{4}} - 1}{y^{\frac{1}{3}} - 1} \right) = ?$

Put $y = x^{12}$

$$= \lim_{x \to 1} \frac{(x-1)(x^2 + 1 + x)}{(x^2 - 1)(x^2 + 1)}$$

$$= \lim_{x \to 1} \frac{(x-1)(x^2 + 1 + x)}{(x-1)(x+1)(x^2+1)} = \frac{1+1+1}{(2)(2)} = \frac{3}{4}$$

$$= \lim_{x \to 1} \left[\frac{x^3 - 1}{x^4 - 1} \right]$$

Q. $\lim_{x \to 1} \left(\frac{x^{\frac{1}{4}} - 1}{x^{\frac{1}{5}} - 1} \right) = ?$

Put $x = y^{20}$

$= \lim_{y \to 1} \left(\frac{y^5 - 1}{y^4 - 1} \right) = ?$ ($\because$ easy to factorise)

$= \frac{0}{0}$ form.

$= \lim_{y \to 1} \left(\frac{5y^4}{4y^3} \right) = \lim_{y \to 1} \left(\frac{5y}{4} \right) = \frac{5}{4}$

Type 3 (Rationalisation) →

Q $\lim_{x\to\infty}\left(\sqrt{x^2+x+1}-\sqrt{x^2+1}\right)=?$ $=(\infty-\infty)$ form

$$=\lim_{x\to\infty}\left(\sqrt{x^2+x+1}-\sqrt{x^2+1}\right)\frac{\left(\sqrt{x^2+x+1}+\sqrt{x^2+1}\right)}{\left(\sqrt{x^2+x+1}+\sqrt{x^2+1}\right)}$$

$$=\lim_{x\to\infty}\left[\frac{(x^2+x+1)-(x^2+1)}{\sqrt{x^2+x+1}+\sqrt{x^2+1}}\right]=\lim_{x\to\infty}\left(\frac{x}{\sqrt{x^2+x+1}+\sqrt{x^2+1}}\right)=\frac{\infty}{\infty}\text{ form.}$$

$$=\lim_{x\to\infty}\left[\frac{x}{x\left\{\sqrt{1+\frac{1}{x}+\frac{1}{x^2}}+\sqrt{1+\frac{1}{x^2}}\right\}}\right]=\frac{1}{\sqrt{1+0+0}+\sqrt{1+0}}=\frac{1}{2}\ \text{Ans}$$

Q. $\lim_{t\to+\infty}\left[\sqrt{t^2+t}-t\right]=?=(\infty-\infty)$ form.

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$=$

HW

$=\frac{1}{2}$

Q. $\lim_{x\to 4}\left[\frac{3-\sqrt{5+x}}{1-\sqrt{5-x}}\right]=?=\frac{0}{0}$ form.

(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) ✓ $-\frac{1}{3}$ (d) 3

(M-I) Using L-Hosp Rule :-

$$\lim_{x\to 4}\frac{0-\frac{1}{2\sqrt{5+x}}(0+1)}{0-\frac{1}{2\sqrt{5-x}}(0-1)}=\lim_{x\to 4}\left(-\frac{\sqrt{5-x}}{\sqrt{5+x}}\right)$$

(M-II) using Rationalisation :-

$=\frac{-1}{3}$

HW

Type IV (IND form Concept) →

$\frac{0}{0}, \frac{\infty}{\infty}$, $0\times\infty$, $\infty-\infty$, 0^0 , ∞^0 , 1^∞

Directly use L-Hospital Rule

use L-Hosp Rule after converting them into one of the 1st two forms.

use log Concept.

Note: $\frac{d}{dx}(x^a) = ax^{a-1}$

$\frac{d}{dx}(a^x) = a^x \log_e a$

$\boxed{\frac{d}{dx}(x^x) = x^x(1+\log x)}$

$\frac{d}{dx}(a^a) = 0$

Note: $y = x^x$

$\log y = \log x^x$

$\log_e y = x \log_e x$

$\frac{d}{dx}(\ln y) = \frac{d}{dx}(x \ln x)$

$\frac{1}{y}\frac{dy}{dx} = x\left(\frac{1}{x}\right) + \ln x(1)$

$\frac{dy}{dx} = y(1+\ln x)$

Q1: $\lim_{x \to a} \left(\frac{x^x - a^a}{x^a - a^x} \right) = ? = \frac{0}{0}$ form $= \lim_{x \to a} \left(\frac{x^x(1 + \log_e x) - 0}{a x^{a-1} - a^x \log_e a} \right)$

$= \frac{a^a(1 + \log_e a)}{a a^{a-1} - a^a \log_e a} = \frac{a^a(1 + \log_e a)}{a^a(1 - \log_e a)} = \frac{1 + \ln a}{1 - \ln a}$ Ans

Q2: $\lim_{x \to \infty} \left(\frac{x^n}{e^x} \right) = ? = \frac{\infty}{\infty}$ form $= \lim_{x \to \infty} \frac{D^n(x^n)}{D^n(e^x)} = \lim_{x \to \infty} \left(\frac{n!}{e^x} \right) = \frac{n!}{e^\infty}$

where n = +ve Integer. Here we are using L-Hosp Rule n times.

$= \frac{n!}{\infty} = 0$ Ans

Note:

$D^n(x^n) = n!$

$D^{n+1}(x^n) = 0$

Detailed Exp :-

$$\lim_{x\to\infty}\left(\frac{x^n}{e^x}\right)=\frac{\infty}{\infty}=\lim_{x\to\infty}\frac{n(x)^{n-1}}{e^x}=\frac{\infty}{\infty}=\lim_{x\to\infty}\frac{n(n-1)x^{n-2}}{e^x}=\frac{\infty}{\infty}$$

$$=\lim_{x\to\infty}\frac{n(n-1)(n-2)x^{n-3}}{e^x}=\frac{\infty}{\infty}=\cdots\cdots=\lim_{x\to\infty}\left(\frac{n(n-1)\cdots 3.2.x^1}{e^x}\right.$$

$$=\frac{\infty}{\infty}=\lim_{x\to\infty}\frac{n!}{e^x}\neq\frac{\infty}{\infty}=\left.\left(\frac{n!}{e^\infty}\right)=\frac{n!}{\infty}=0\right)$$

Note:

$D(x) = 1 = 1!$

$D^2(x^2) = D(2x) = 2 \times 1 = 2!$

$D^3(x^3) = D^2(3x^2) = D(3 \times 2x) = 3 \times 2 \times 1 = 3!$

$D^4(x^4) = D^3(4x^3) = D^2(4 \times 3x^2) = D(4 \times 3 \times 2x) = 4 \times 3 \times 2 \times 1 = 4!$

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In general $\boxed{D^n(x^n) = n!}$ & $D^{n+1}(x^n) = D\left(D^n(x^n)\right)$

$= D(n!) = 0$

ie $\boxed{D^{n+1}(x^n) = 0}$

THANK - YOU

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